Gate A: deployed component mesh

Progress report, system-test record and show-and-tell

Evidence field	Recorded value
Status	Closed; founder accepted and authorised publication on 1 September 2026
Walkthrough	1 September 2026, local synthetic Minikube environment
Source revision	`82d5ff3f27f61a04ad045899c838f20e2d9e258e`
Image digest	`sha256:b71786a0ad7fb59186ef9c58d53915a1cb5178de69b277fb9f445d1dd4b66639`
Environment	`environment-c5ef1307484826d1b598ad73a08d31b6`
Cluster profile	`public-purpose-lab-gate-a`
Trust profile	Environment-local synthetic root; synthetic data only
Canonical requirement	`docs/scenarios/demonstration-scenarios-and-functional-requirements.md`

> Gate A was accepted and closed by the founder on 1 September 2026. Closure accepts the bounded component-mesh evidence; it does not make a production claim or establish legal, regulatory, clinical or compliance assurance.

Outcome at a glance

Gate A turns the approved component list into a real deployed mesh. Twelve separately identified workloads now run in Kubernetes, authenticate to the event infrastructure, report readiness and appear in a new Operations Console. The nine configurable component skeletons accept a bounded, non-business capability probe and return conclusive outcomes under one correlation identifier.

The walkthrough also re-opened the inherited Director, Presentation and Workbench surfaces and repeated the mesh test after restarting NATS and all nine configurable workloads. No source intake, knowledge ingestion, review decision or report generation has been simulated. Those user-visible capabilities remain in later gates.

Approved objective and acceptance position

The approved Gate A objective was to deploy all required skeleton instances, authenticate them, publish honest readiness and make them visible in operations views. The accepted architecture allowed one configurable Rust component-host binary while preserving separate workload identities, permissions, readiness and failure boundaries.

Acceptance requirement	Evidence	Position
Twelve expected components report ready	Operations screen and `/api/v1/mesh` showed 12 of 12	Achieved
Twelve distinct workload identities	Automated smoke checked uniqueness; the screen exposes each public NKey identity	Achieved
Operations refreshes from real events	Missing/stale/ready is derived from observed heartbeats, not seeded ready state	Achieved
Nine bounded commands have conclusive correlated outcomes	One probe issued nine commands; nine `component.command-accepted` outcomes were visible	Achieved
Broker and component restart recover safely	NATS and all nine component hosts were restarted; post-restart smoke passed	Achieved
Existing surfaces remain usable	Director, Presentation and Workbench were opened from the same image	Achieved
Screenshot-backed report and verified PDF	This canonical report, its six screenshots and its rendered PDF	Achieved for review
Founder accepts the implemented flow	Founder accepted closure and publication on 1 September 2026	Achieved

What existed before Gate A

M1-M3.4 already provided the architecture catalogue, versioned contracts, authenticated NATS carriage, environment-scoped synthetic identity, a scenario Director, presentation gateway, common Workbench surface, local Minikube deployment and a managed Google Cloud preview lifecycle. Those foundations proved identity and presentation-control invariants, but they did not deploy the complete source-to-report component set or provide a component-wide operational view.

Gate A retains those foundations and adds only the minimum mesh needed for visible, testable progress. The existing `scenario-director`, `presentation-gateway` and `identity-broker` continue to own CTL-01, CTL-02 and IAM-01. Nine new deployments use a shared configurable host image for their currently bounded responsibilities.

Demonstrated flow

Step	Actor and screen	Command or event path	Owning behaviour	Visible result
1	Operator starts the dedicated Minikube profile	Kubernetes starts NATS and 12 workloads	Each workload has its own configuration, identity and health boundary	12 of 12 pods become ready
2	Presenter opens `OPS-COMPONENTS`	Readiness heartbeats arrive on ppl.gate-a.events.<component>	OPS-01 projects only actually observed readiness	12 cards show ready with distinct instance and workload IDs
3	Presenter selects **Probe component mesh**	OPS-01 publishes nine `O-001` v0.1 capability commands	Each target validates version, issuer, environment, purpose, target and idempotency	One correlation ID is shown
4	Components conclude the probe	Each target publishes accepted, duplicate or refused outcome	The probe changes no business state	Nine accepted outcomes appear together in `OPS-EVENTS`
5	Presenter opens Director, Presentation and Workbench	Existing HTTP surfaces are served by the exact same image	Existing M3 responsibilities remain separate from Gate A probes	All three inherited surfaces remain reachable
6	Operator restarts NATS and all component hosts	Heartbeats cease and then resume; in-memory projection rebuilds	Readiness is recovered from new observations	12 ready and nine outcomes pass again after restart

```
Director (CTL-01) -----------+
Presentation (CTL-02) -------+--> authenticated readiness events --+
Identity (IAM-01) -----------+                                      |
                                                                    v
AUT / DOM / CNT / KNO / WRK / RPT / AUD / OPS / INT instances --> NATS
       ^                                                            |
       +---------- bounded capability commands from OPS-01 <--------+
                                                                    |
Operations Console <------ real readiness and outcome projection <--+
```

Walkthrough evidence

1. Complete mesh readiness

The Operations Console reports 12 of 12 ready for the exact environment. The status is calculated from component-owned readiness events. An expected catalogue may identify a missing component, but it cannot manufacture a ready result.

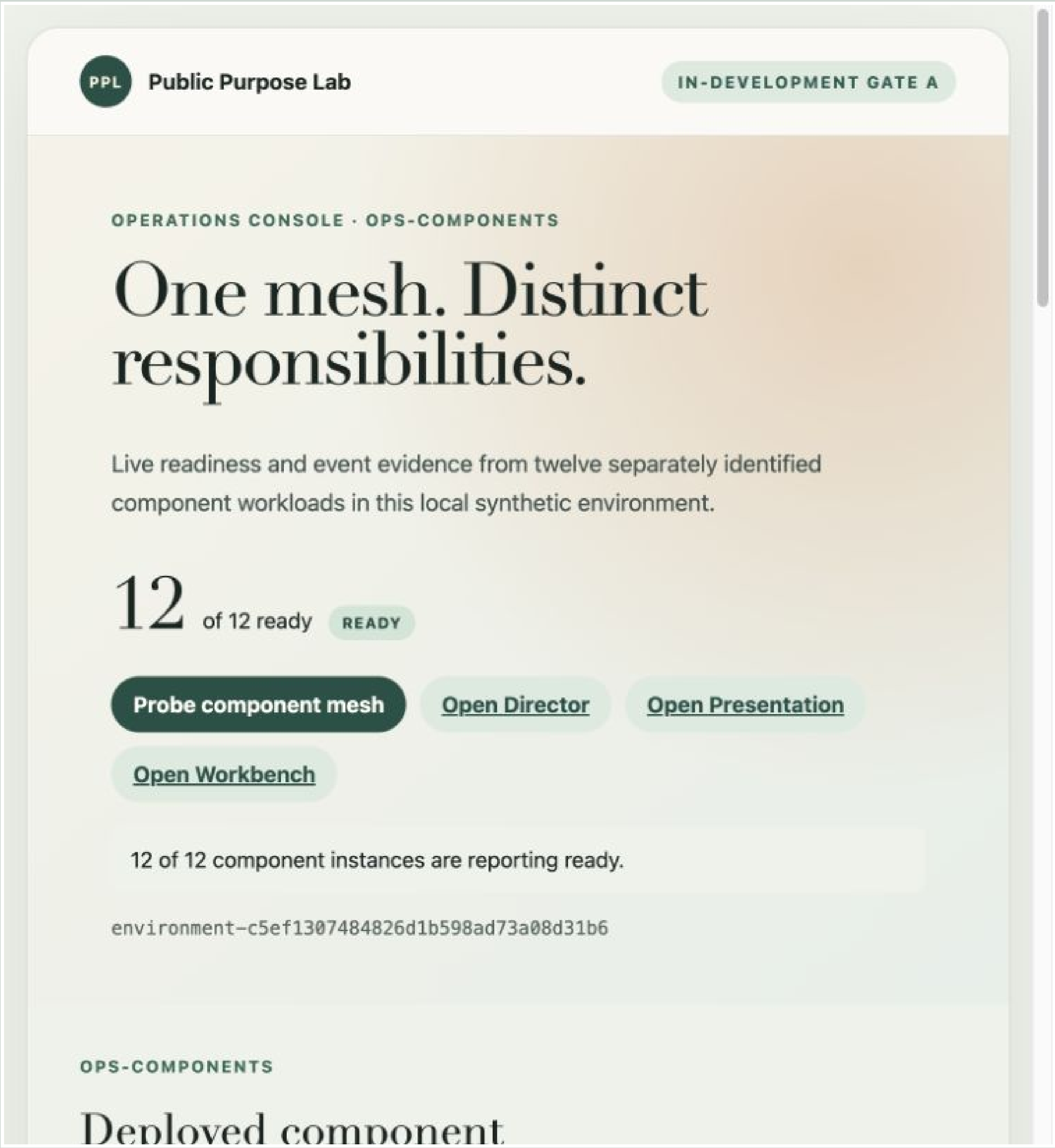

Figure 1. OPS-COMPONENTS showing 12 of 12 workloads ready in the evidenced environment.

2. One bounded correlated probe

The presenter issues one visible probe. The UI records that nine bounded capability commands were sent under correlation `correlation:de7687eb-df24-42f0-bec0-edbd757b713e`.

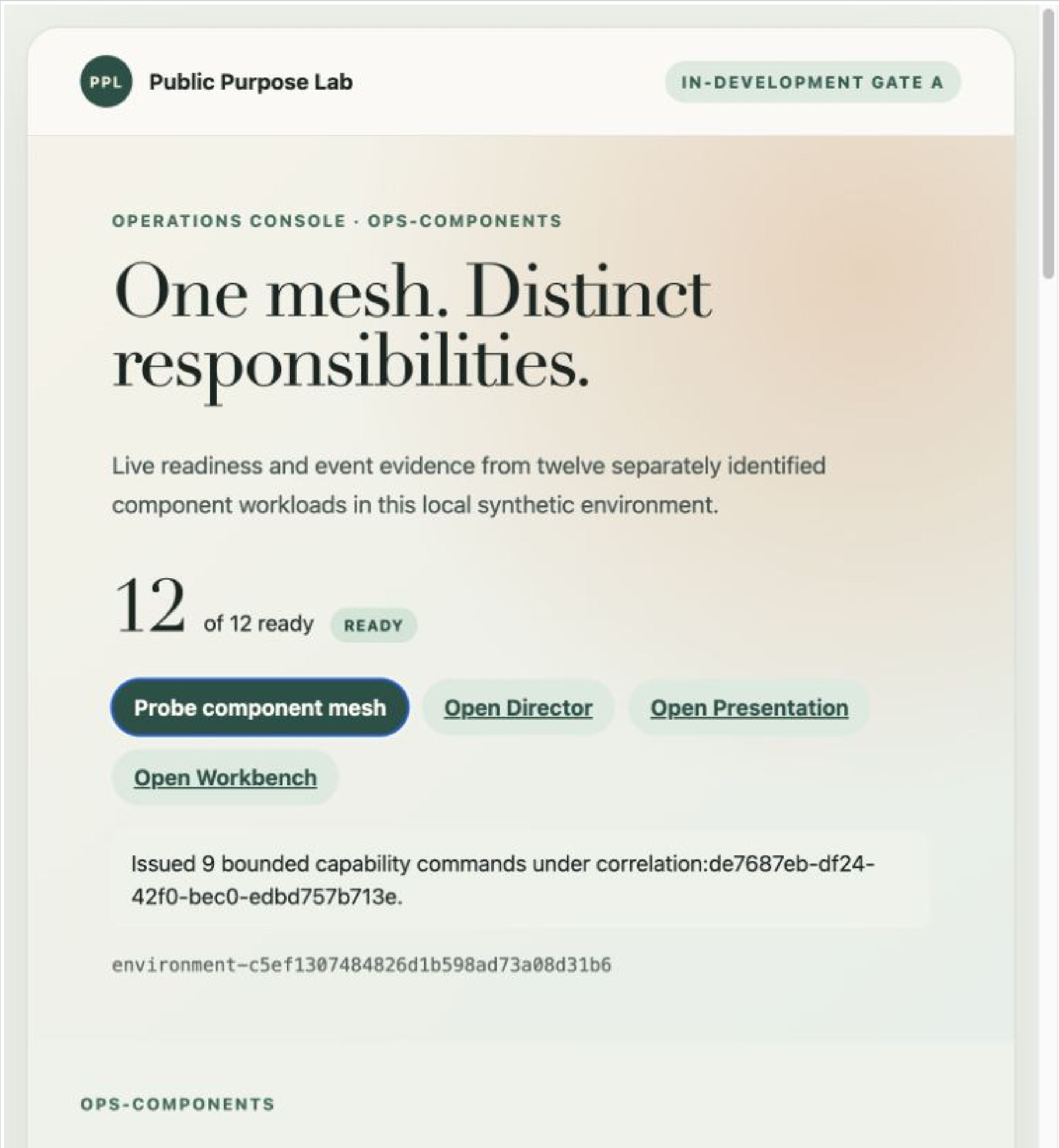

Figure 2. The issued probe and its correlation identifier, with the mesh still reporting ready.

3. Conclusive component outcomes

OPS-EVENTS shows the nine accepted outcomes for AUT-01, DOM-01, CNT-01, KNO-01, WRK-01, RPT-01, AUD-01, OPS-01 and INT-01. Readiness heartbeats are hidden by default so the conclusive outcomes remain legible.

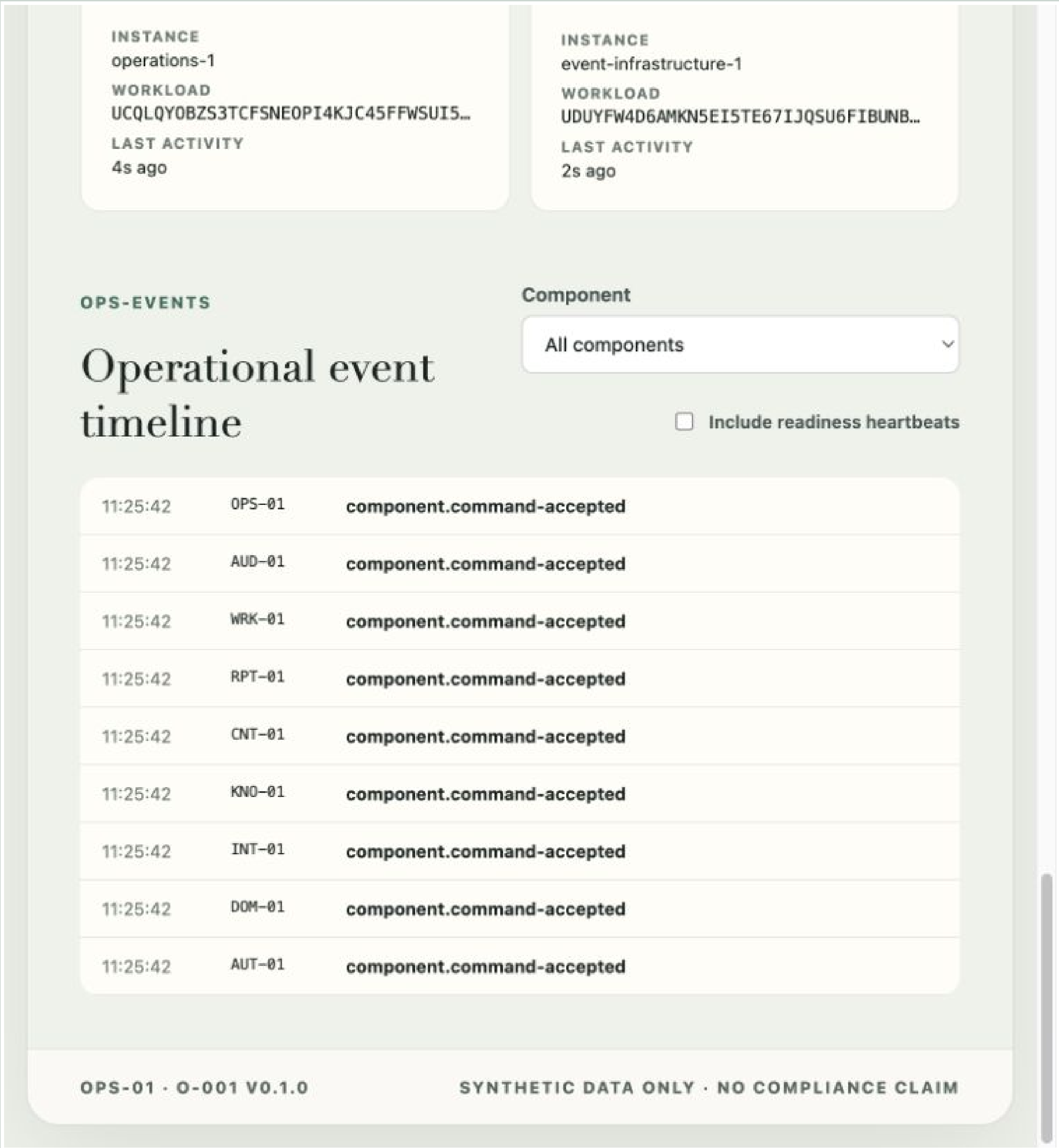

Figure 3. All nine `component.command-accepted` outcomes visible together after the probe.

4. Inherited Director surface

The Director remains the scenario-control surface. Gate A does not add scenario catalogue, environment or run controls; the current screen continues to label itself as an in-development walking skeleton.

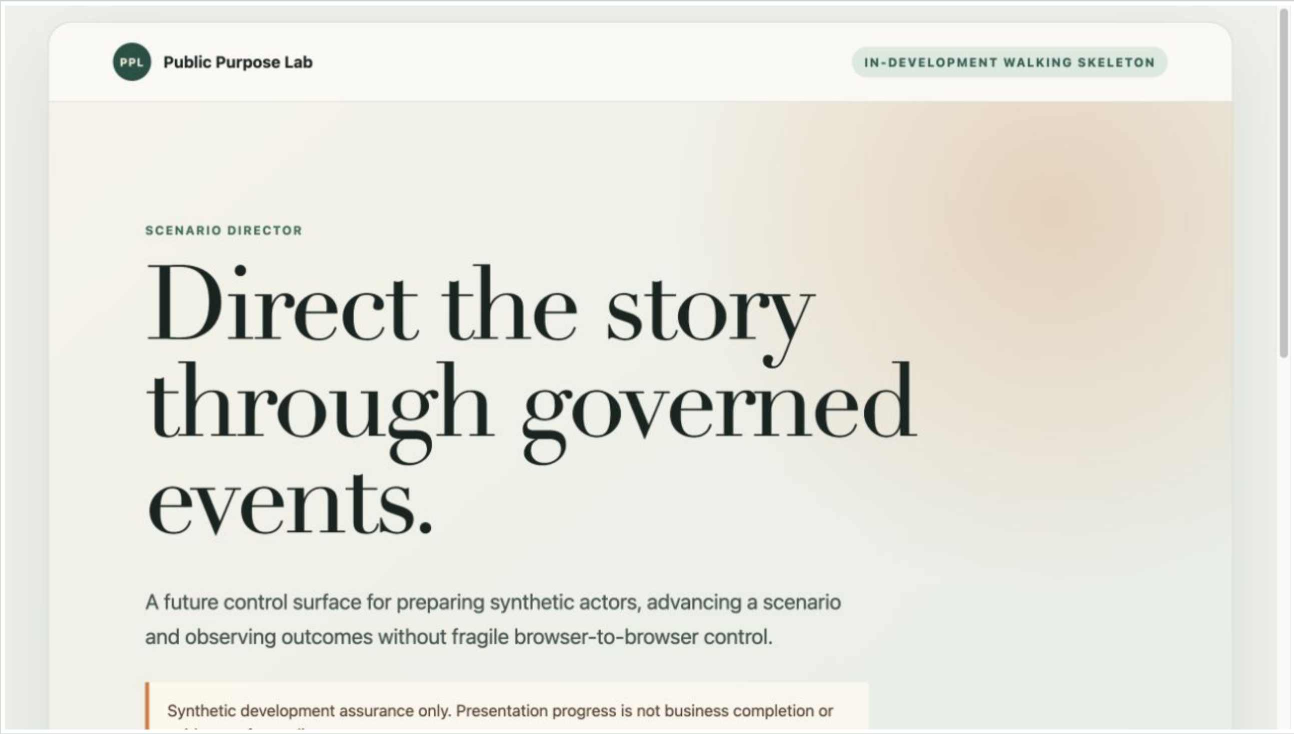

Figure 4. Existing Director surface reopened from the evidenced Gate A image.

5. Inherited Presentation surface

The Presentation surface remains independently addressable and target-owned. Gate A does not claim that a component probe is a presentation or business outcome.

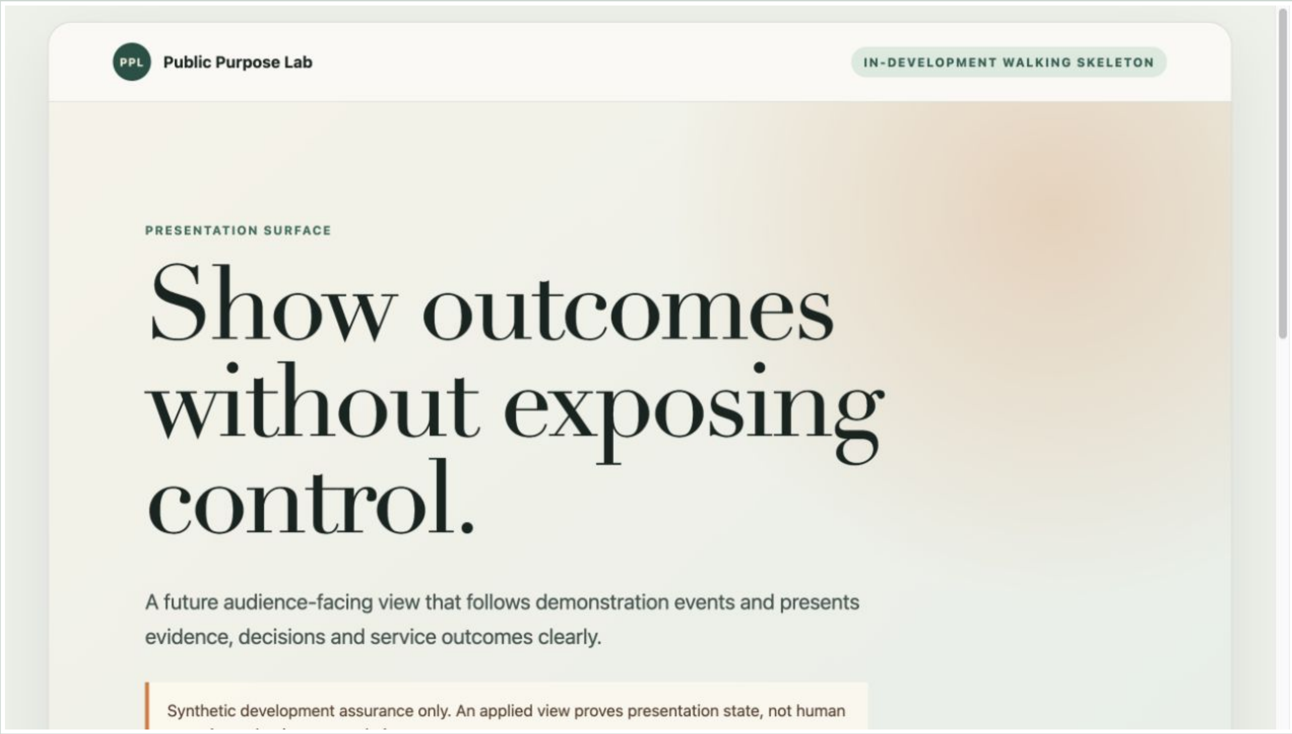

Figure 5. Existing Presentation surface reopened from the evidenced Gate A image.

6. Inherited Workbench surface

The Workbench remains a governed surface skeleton. Upload, paste, source status, knowledge query, review and report functions are intentionally not introduced in Gate A.

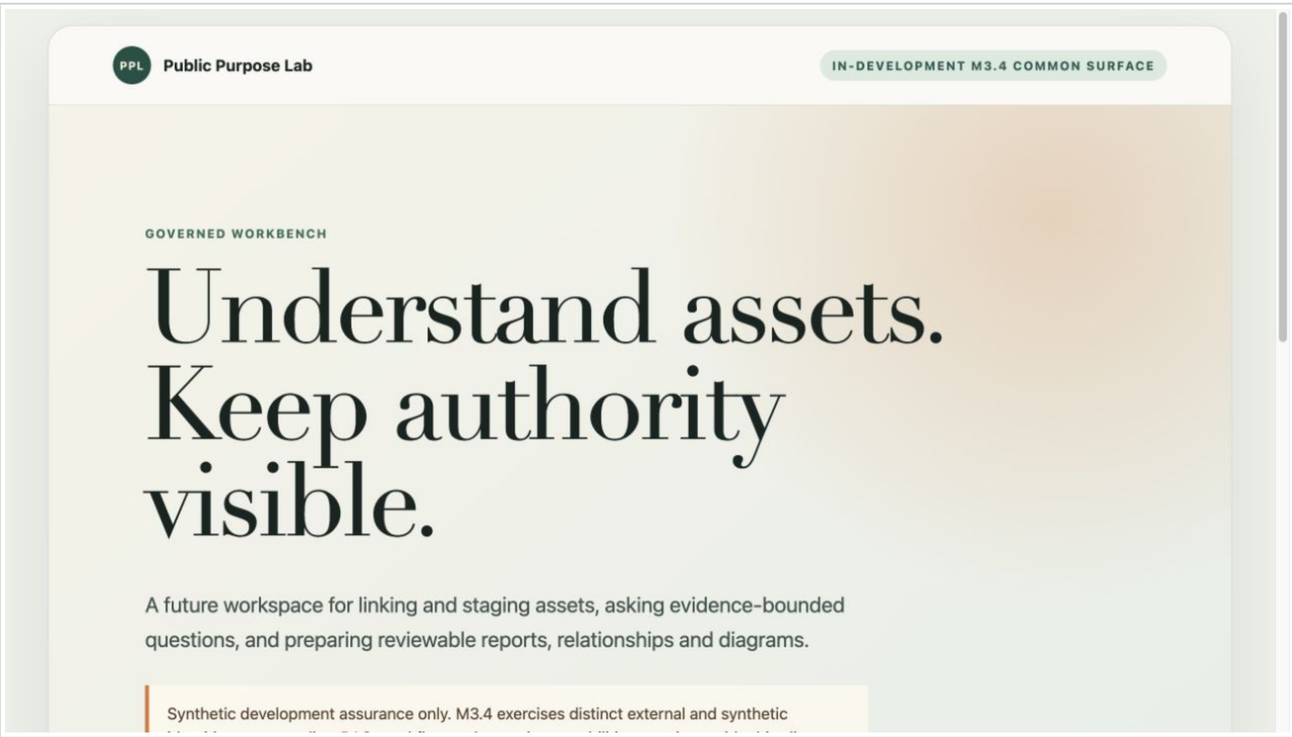

Figure 6. Existing Workbench surface reopened from the evidenced Gate A image.

Components and responsibilities exercised

Component	Deployed instance	Gate A behaviour exercised	Business behaviour deliberately absent
CTL-01	`scenario-director-1`	Authenticated readiness; inherited M3 flow remains available	New scenario catalogue or orchestration
CTL-02	`presentation-gateway-1`	Authenticated readiness; inherited presentation surface remains available	New semantic Gate B views
IAM-01	`identity-broker-1`	Authenticated readiness; inherited synthetic identity flow passes	New Workbench actor placement UI
AUT-01	`authorisation-1`	Bounded capability command and conclusive outcome	Protected business-action decision
DOM-01	`engagement-1`	Bounded capability command and conclusive outcome	Engagement creation
CNT-01	`source-governance-1`	Bounded capability command and conclusive outcome	Upload, quarantine, validation or staging
KNO-01	`knowledge-processing-1`	Bounded capability command and conclusive outcome	Ingestion, RAG or cited query
WRK-01	`review-workflow-1`	Bounded capability command and conclusive outcome	Review task or human decision
RPT-01	`reporting-1`	Bounded capability command and conclusive outcome	Report preview or release
AUD-01	`audit-evidence-1`	Bounded capability command and conclusive outcome	Durable scenario reconstruction
OPS-01	`operations-1`	Real readiness projection, probe issue and event timeline	Durable logs or evidence store
INT-01	`event-infrastructure-1`	Authenticated event carriage and capability outcome	Business ownership or exactly-once claim

Screens and functions added or changed

Surface	Added or changed	Function and rule
`OPS-COMPONENTS`	New Operations Console	Shows expected instances against observed readiness, source revision, image digest, public workload identity and last activity
`OPS-EVENTS`	New event timeline	Shows privacy-minimised event type, component and time; readiness heartbeats are optional
Component probe control	New bounded action	Issues one no-business-effect command to each of the nine configurable instances under one correlation ID
Director	Existing surface retained	Link from Operations; no new Gate A scenario claim
Presentation	Existing surface retained	Link from Operations; presentation state remains distinct from business completion
Workbench	Existing surface retained	Link from Operations; later business functions remain clearly future work

Contracts and rules exercised

The Gate A `O-001` v0.1.0 event shape is a working implementation binding, not an approved canonical logical contract. Every command carries contract and version, issuer, target, environment, purpose, correlation, causation, idempotency and time. Every receiver validates those fields before accepting the probe.

- Each workload uses an environment-generated NKey and least-privilege NATS publish/subscribe permissions.
- Component ID, instance ID and workload identity are distinct and visible.
- `ready` is derived from an observed heartbeat no more than 15 seconds old; unobserved is `missing`, expired is `stale`.
- Exact command redelivery returns a duplicate outcome; changed content under the same idempotency key is refused.
- Capability probes have no business side effect and cannot create engagement, source, knowledge, review or report state.
- Operations events exclude credentials, private keys and source bodies.
- The environment-local synthetic root remains inside the environment setup directory and is not mounted as an application private key.
- The managed-hosted overlay does not enable local-synthetic Gate A readiness by default.

Every configurable host exposes liveness, readiness, understood-contract and component-summary endpoints. OPS-01 additionally exposes mesh, event and probe endpoints. These endpoints support Gate A observability; they are not a general management API.

System-test evidence

Check	Exact evidenced result
Repository quality suite	Architecture 21 components/40 contract families; 20 schemas/40 fixtures/20 compatibility descriptors; formatting, links, frontend types/tests, Rust formatting/clippy/tests, M1 and M2 runtime checks all passed
Rust tests	60 tests passed, including two new component-host tests
Gate A cold smoke	12 distinct workloads ready; nine conclusive outcomes under one correlation
Inherited M3 smoke	External/synthetic session path, CSRF refusal, semantic cue, checkpoint, reset and successor isolation passed
Restart smoke	NATS plus all nine configurable workloads restarted; 12 ready and nine outcomes passed again
Kubernetes manifests	Gate A Helm lint/template and both M3 Kustomize variants rendered successfully
Shell and YAML checks	Changed shell scripts passed ShellCheck; Compose parsed as 14 services; `git diff --check` passed

The final post-restart walkthrough used correlation `correlation:de7687eb-df24-42f0-bec0-edbd757b713e`. The Operations API returned exactly nine accepted outcomes, one from each configurable component.

Adverse evidence and operational lesson

During the exact-revision redeployment, the dedicated cluster was briefly supplied with a newly generated environment identity while its retained IAM volumes still belonged to the original Gate A environment. IAM-01 stopped safely with `IAM state is inconsistent`; it did not silently adopt the new trust domain. The old state and both environment directories were preserved. Reapplying the original `.local/gate-a-minikube` environment restored the consistent trust and state binding.

This is positive fail-closed security evidence, but it also exposes an operator-experience gap. A later change should bind profile, environment ID and retained state explicitly and refuse the rollout before workloads are replaced when they do not match. No retained state should be deleted or repaired implicitly.

Limitations and deferred work

- The nine component instances are genuine deployments and event participants, but their only Gate A handler is a non-business capability probe.
- The Operations projection is in memory and reconstructs readiness from recurring events. It is not durable audit evidence.
- Probe idempotency is process-local because the probe has no side effect. State-changing commands will require durable ownership and reconciliation.
- `O-001` v0.1.0 is a working binding and may change when later flows supply better evidence.
- Missing and stale states are implemented, but the founder walkthrough concentrated on ready, outcome and restart recovery rather than a screenshot of every adverse display state.
- Gate A was exercised in a dedicated local Minikube profile. Compose is represented in CI but was not run through a local Docker daemon, and the Gate A extension was not deployed to Google Cloud.
- Publication was authorised after founder review. Hosted CI and pull-request evidence belong to the repository publication record rather than this local system-test walkthrough.
- No real or confidential data was used. The evidence does not establish production fitness, privacy compliance or accountable business authority.

What Gate A establishes

Gate A establishes a concrete, observable platform shape on which real scenarios can now be built: all named components exist as separately authenticated workloads; they can exchange bounded events; the Operations Console reports what was actually observed; and the inherited presentation-control skeleton remains available.

It does not establish the value path the founders ultimately want. There is still no real scenario catalogue, no visible placement of a synthetic reviewer into Workbench, no source upload, no processing, no cited query, no human review and no report. Gate A was accepted as a mesh gate, not as business-value completion.

Next gate: DS-01 and DS-02

Gate B should make the platform demonstrably useful to a presenter before source processing begins. Its user-visible outcome is a Director-led environment and scenario introduction that places a synthetic reviewer into Workbench and moves the Presentation and Workbench surfaces through semantic views without URL or DOM control.

The founder-review package for Gate B should exercise:

1. `DIR-ENVIRONMENT` with environment, trust profile and honest component readiness;
2. Director external sign-in and `DIR-CATALOGUE` admission reasons;
3. creation of one Demonstration Session without prematurely creating a synthetic application session;
4. backend-only placement and termination of `synthetic-reviewer` in Workbench;
5. `PRES-INTRO`, `WB-ENGAGEMENT` and `WB-SOURCE-INTAKE` semantic view requests;
6. equivalent ordinary Workbench navigation and a visible pause state; and
7. explicit refusal of unsupported or expired view requests.

Gate B should select only the contract bindings required by that walkthrough. DS-03 source intake remains the following gate and the first functional business-processing path.